BioSysteM Research Retreat in Ettal

My road to founding and agentic bioengineering

by Tim Kucera

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Founder trajectory

Wet lab, engineering, and modeling in search for programmable biology



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Biology

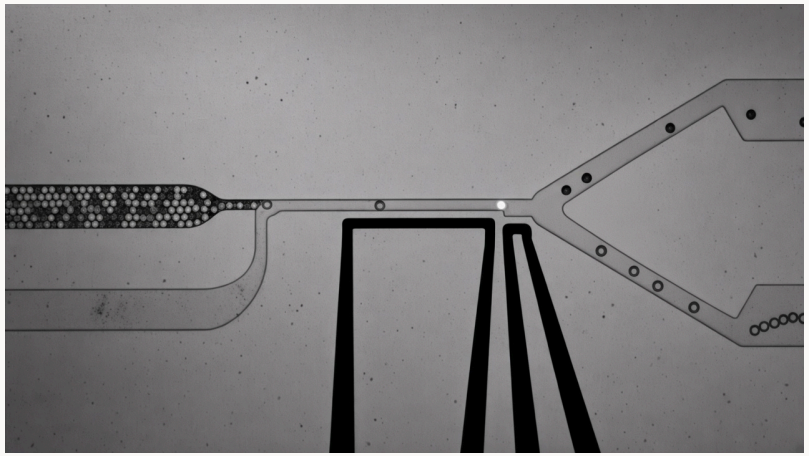
MSc
ETH Zürich
Biotechnology

2021
DropSort
Droplet Sorting Software

PhD
ETH / MPI
Molecular Machine Learning

2025
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AI infrastructure for Biotech

2026
Accubator
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High-throughput droplet sorting on a microfluidic chip

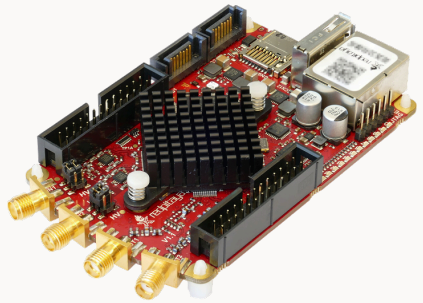
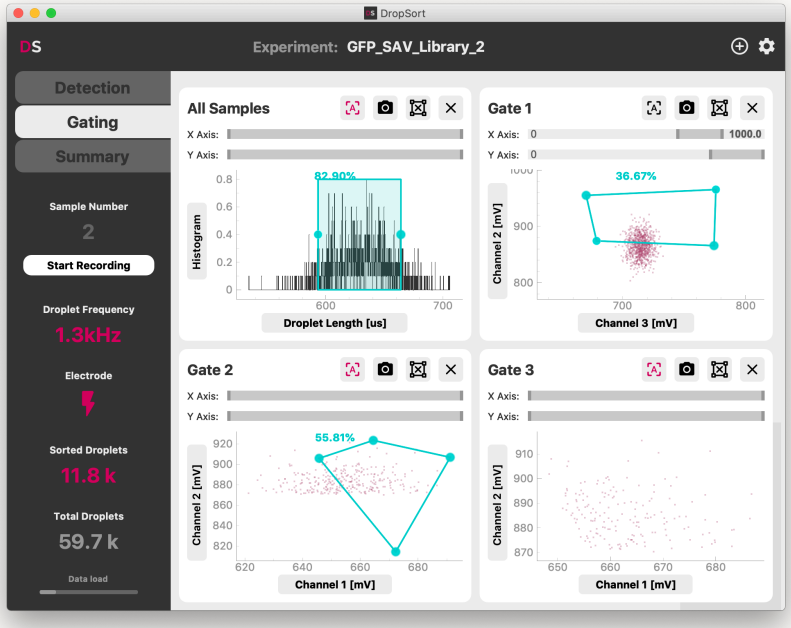


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A high-performance data processing card (FPGA)



DropSort software for running FPGA-based fluorescence activated droplet sorting.

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Image adapted from Goodfellow et al. (2014)

Generative modeling for protein design and engineering

ProteinShake

Kucera et al. (2023)

bioverse

Kucera et al. (unpublished)

Frameworks for standardizing biological data for machine learning analyses

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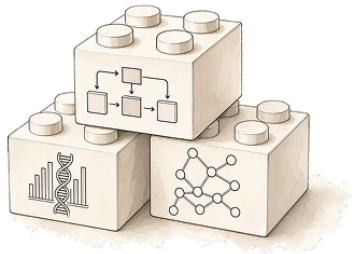
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Public databases,
papers, models

The Imaginary Platform



Standardized biological **data**, machine learning **models**,
and processing **workflows**



ML Engineers
Tools for developing models
Accelerating development cycles



Biotech Researchers
Graphical user interfaces
Accessible models for non-technical users

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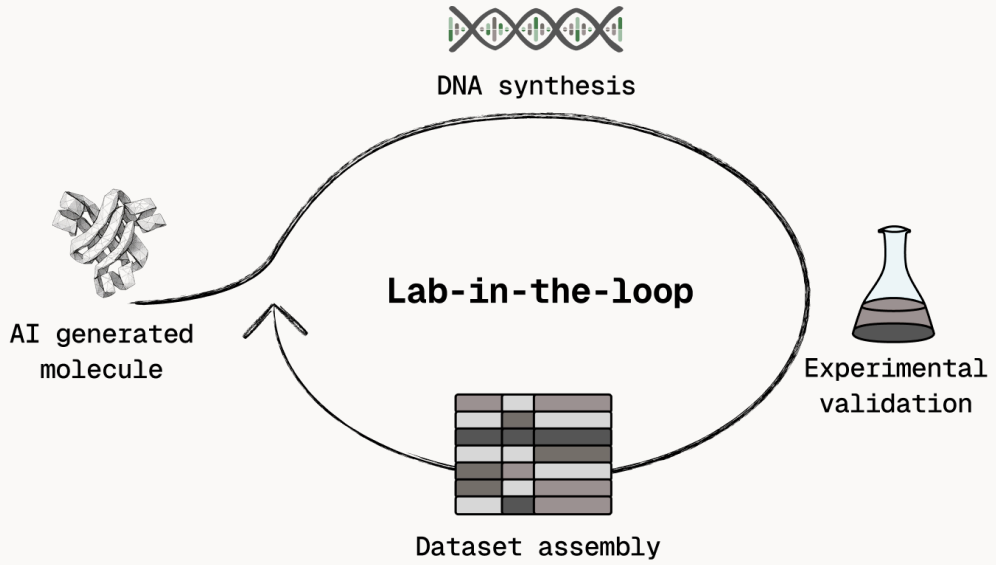
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More information at: www.bio-m.org/accubator

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Thinking about a company? Just do it!

Much of what you are doing as a researcher has a market value

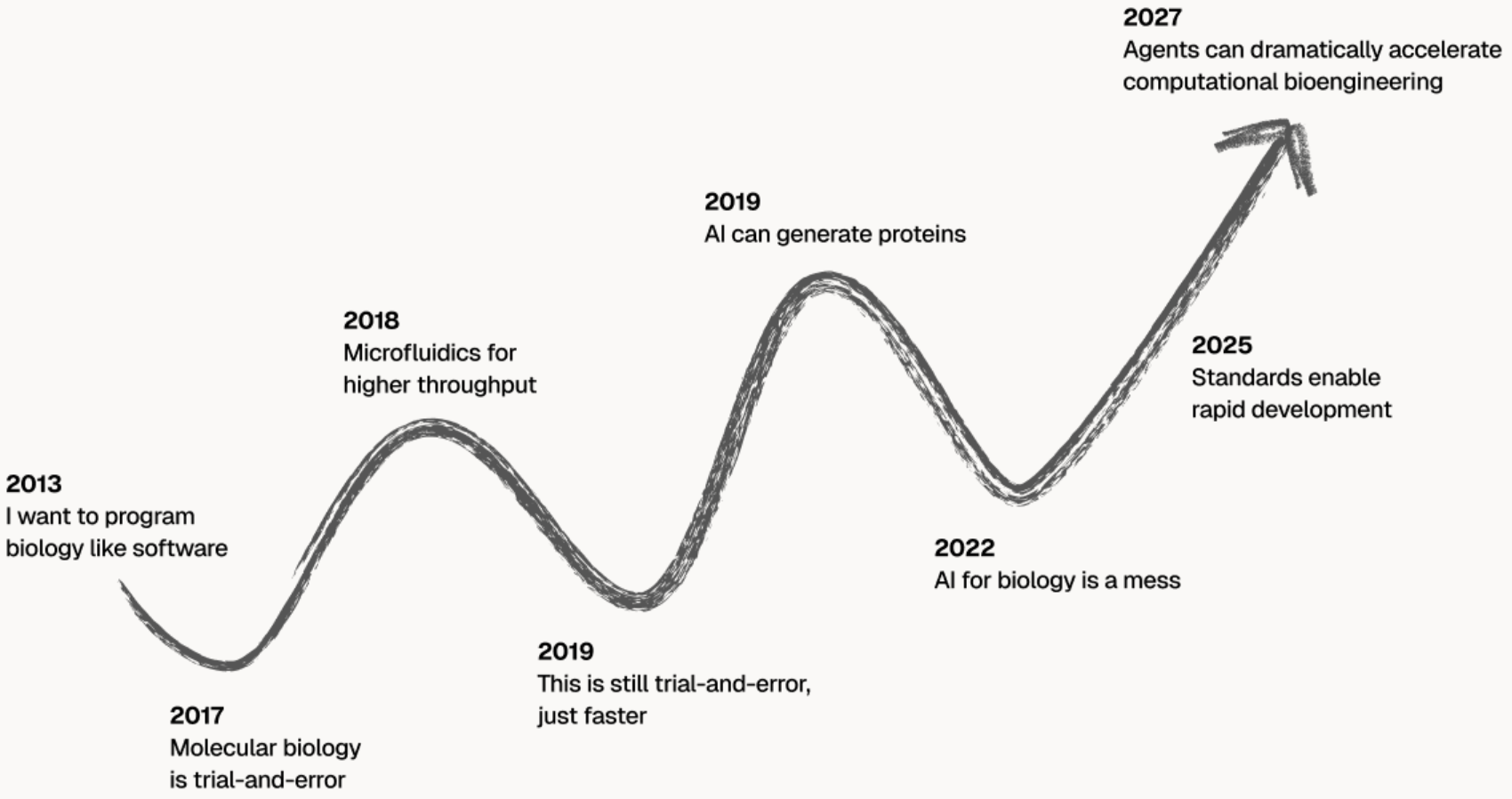
Founding is easy and does not always need much money or time

Be open to opportunities and they will come

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Where are we now with programmable biology?

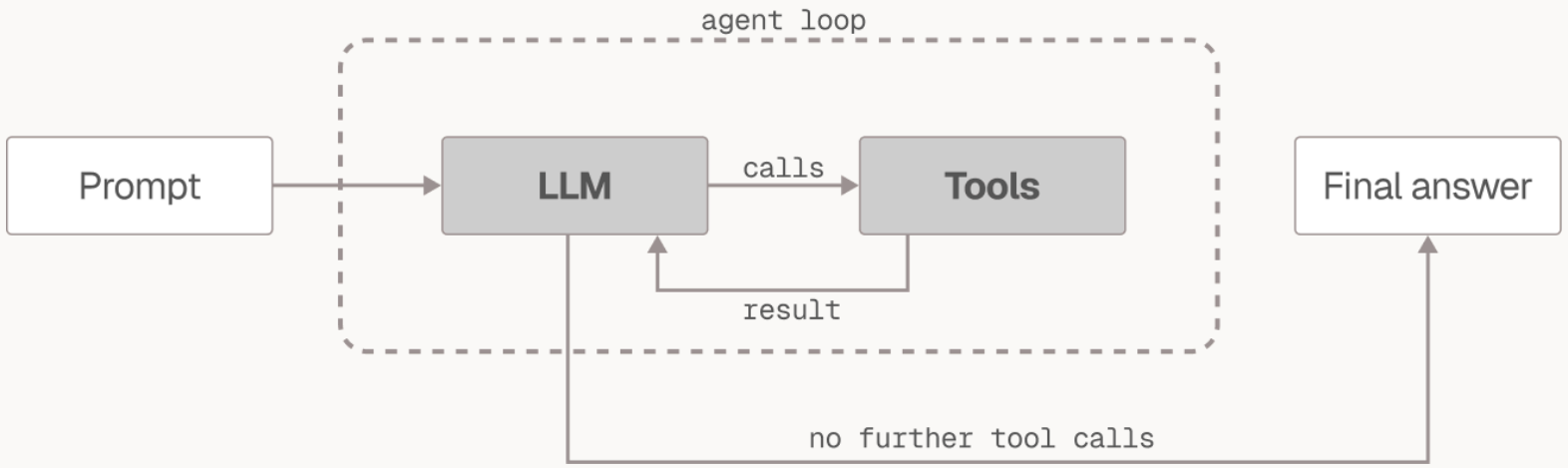
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The agent loop

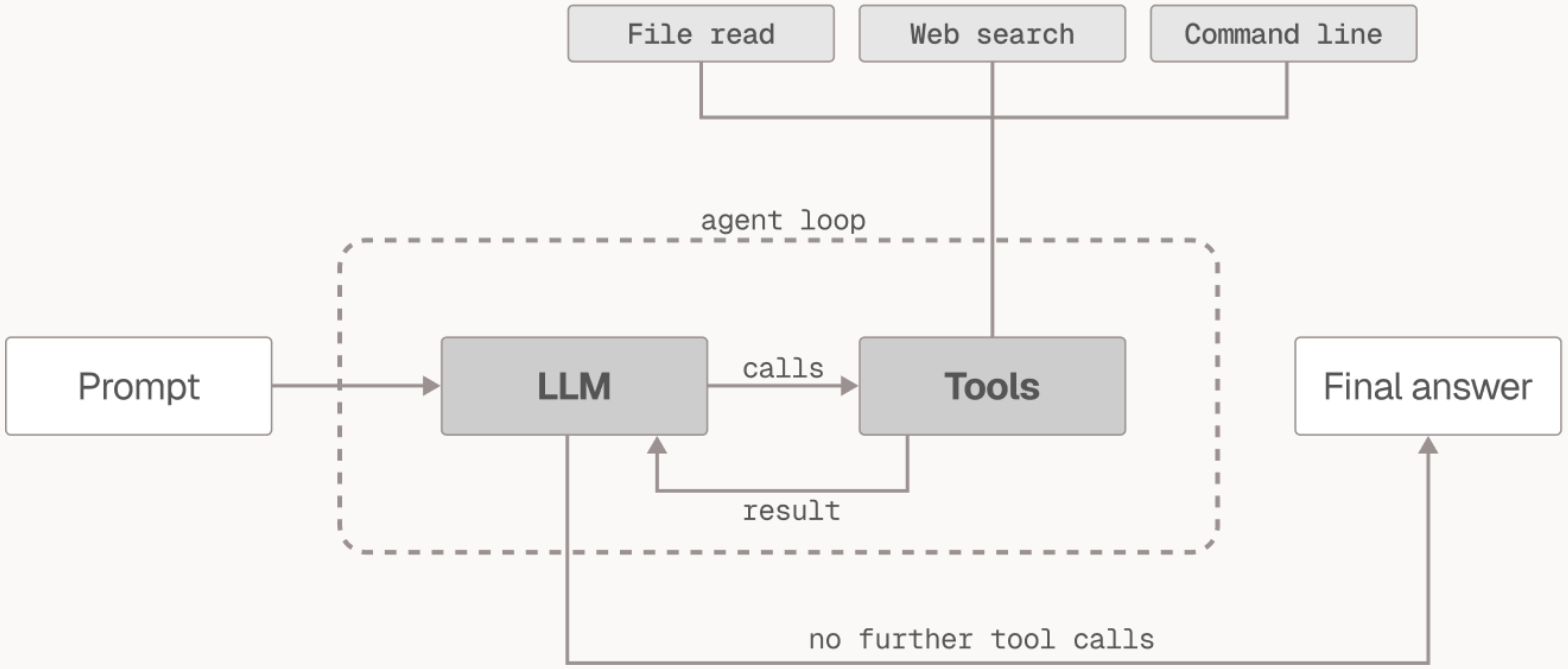
An LLM with tool calls, iterating until a goal is met



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The agent loop

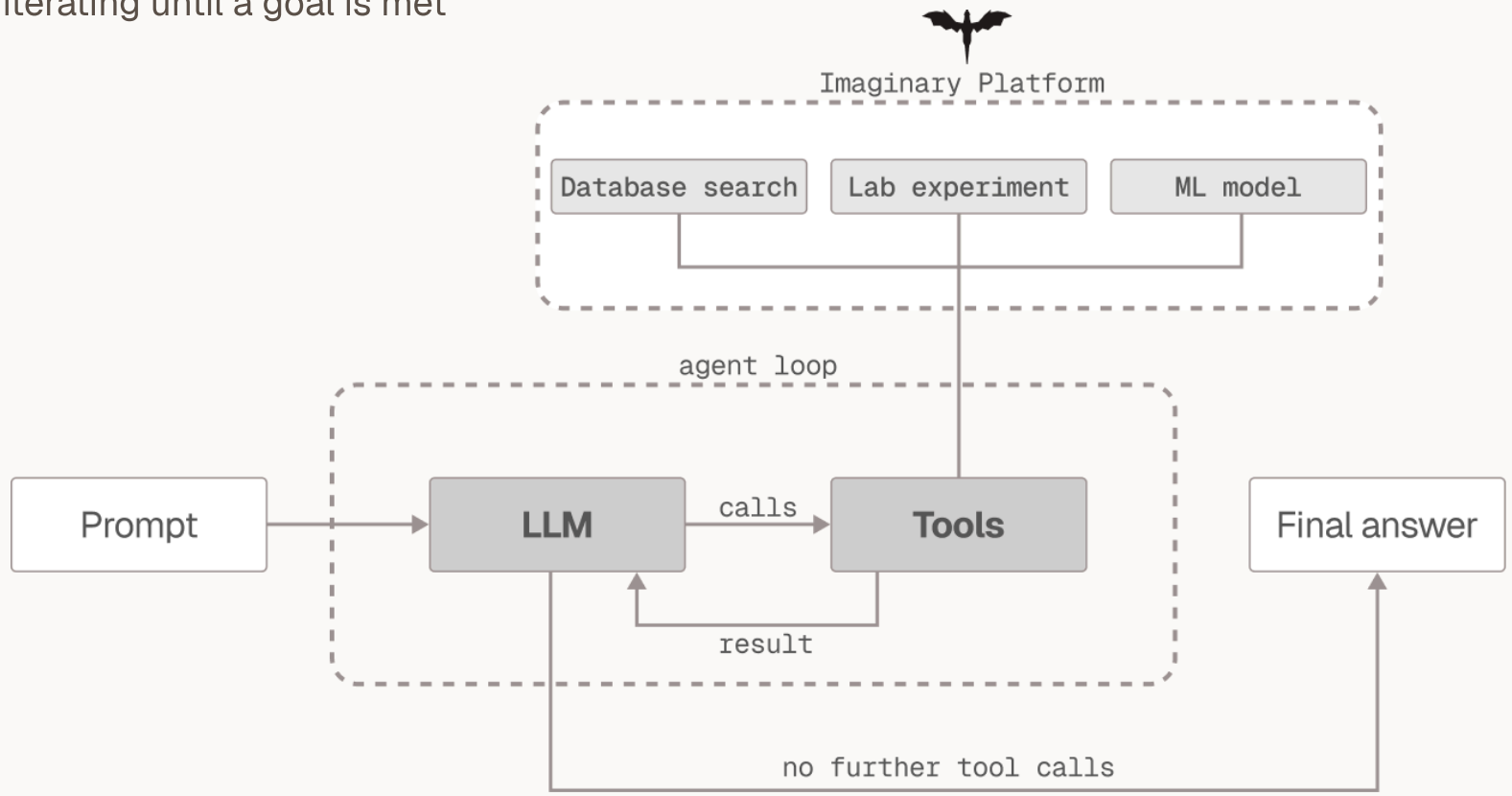
An LLM with tool calls, iterating until a goal is met



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The agent loop for bioengineering

An LLM with tool calls, iterating until a goal is met



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What becomes possible

Natural-language prompts on real infrastructure

We created this binder, will it have off-targets in our model organism? How much can we trust your prediction?

binder.pdb

Turn our DMS assay results into a dataset and train a model to predict the next mutations to test.

dms_assay.csv

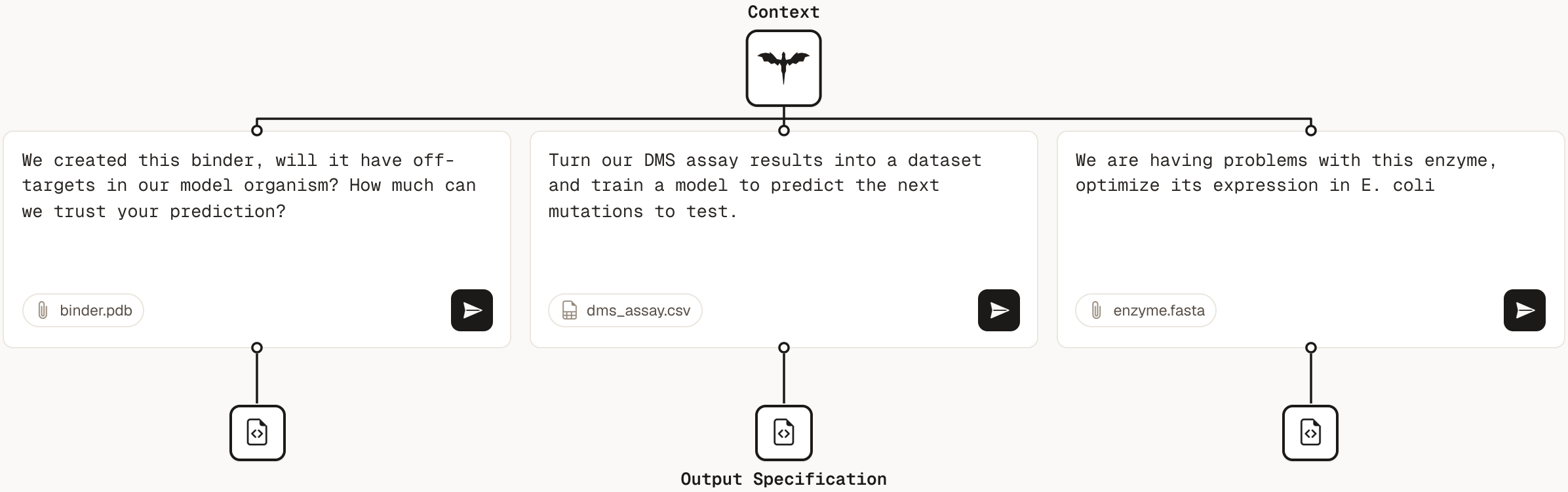
We are having problems with this enzyme, optimize its expression in E. coli

enzyme.fasta

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What becomes possible

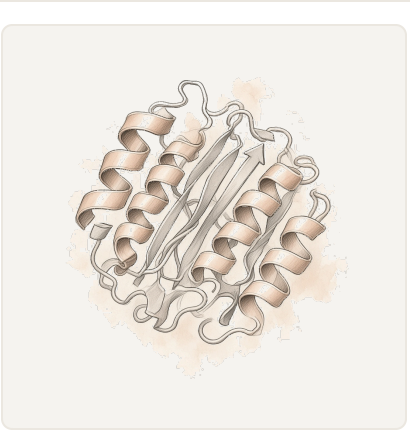
Natural-language prompts on real infrastructure



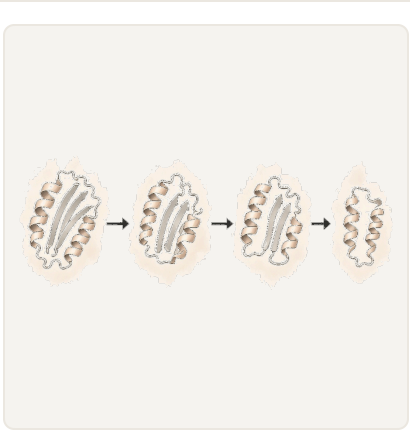
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The long-term vision for Imaginary

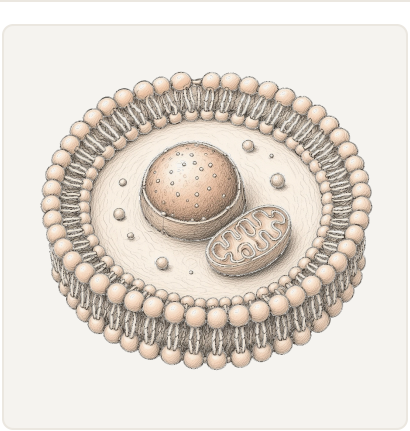
From the molecular level to computational design of biological systems



Molecules



Systems of molecules



Synthetic cells



Multicellular synthetic organisms

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SYSTEMS BIOLOGY

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